

From Databricks Meets PyTorch to Databricks Meets Keras

Two architectural patterns for enterprise AI — research flexibility vs operational simplicity — and how **ASRA** implements both on the path to scientific intelligence.

Databricks + PyTorch = Infrastructure + Research Flexibility

Databricks + Keras = Infrastructure + Simplicity

DATABRICKS MEETS PYTORCH

Enterprise data/compute + maximum model development control.

- Lakehouse, Spark pipelines, features, MLOps
- Custom architectures, RL, world models, LLMs
- Frontier research — build anything

"Unify data and training in one environment."

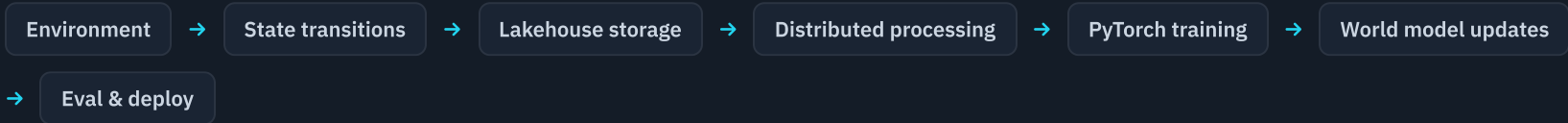
DATABRICKS MEETS KERAS

Same infrastructure + radical ease of use.

- Abstracted pipelines, tracking, deployment
- `model.fit(data)` — complexity hidden
- Domain experts focus on problems, not clusters

"A full AI factory that feels like a workbench."

UNIFIED PLATFORM FLOW (ASRA EXAMPLE)



WHO IS THE PLATFORM FOR?

Aspect	+ PyTorch	+ Keras
Primary user	ML researchers, engineers	Scientists, domain experts
Goal	Maximum flexibility	Maximum productivity
Learning curve	Steep	Easier
Novel architectures	Ideal	More constrained
Time to first model	Slower	Faster
World models	Ideal	Possible, less flexible

ASRA DUAL-LANE ARCHITECTURE

Research	PyTorch-class · <code>asra-arc/src/asra/</code> · compose cognitive phases, invent mechanisms
Delivery	Keras-class · <code>kaggle-notebooks/phase*/</code> · versioned agents, simple submit surface
Science	Intelligence platform · Phases 4–8 + Decision Biology · hypothesize → experiment → refine

Don't collapse lanes too early — library needs competition validation; agents need library depth.

ASRA maturity path

- ① Databricks + PyTorch — build the technology
- ② Databricks + Keras — make it accessible
- ③ Scientific Operating System — continuous discovery at scale

